Discharge Compass

Grounded, verified, multilingual after-hospital care plans | **Theme:** HealthTech (with AI) | 2nd LaserHack 2026, Round 1

Team name	The Legends
Members (2)	Parv Tiwari (Team Lead), Parth Gupta
Contact	parvtiwari0111@gmail.com, 9285121000 | Shri Govindram Seksaria Institute of Technology and Science, Indore

1. Problem Statement and Theme

Theme: HealthTech. The first days after leaving a hospital are among the riskiest in healthcare. Patients go home with a dense discharge summary full of jargon, new or changed medicines, follow-up appointments and warning signs. Historically about one in five Medicare patients has been readmitted within 30 days, and many of these returns are avoidable. Medication confusion, missed follow-ups and not knowing when to call a doctor are leading causes.

Who is affected: patients and family caregivers, especially older adults, people with limited health literacy, and non-native English speakers, plus overstretched nurses who have only minutes to explain everything at discharge. Existing patient portals simply show the same dense document again. Generic AI chatbots can summarize it, but they can hallucinate, and a wrong medical instruction is worse than none.

2. Proposed Solution

Discharge Compass turns a discharge summary into a personal, plain-language recovery plan that is **verified, not just summarized**. A patient or caregiver uploads or photographs the summary. The system extracts structured data (medicines, doses, appointments, restrictions, red-flag symptoms), rewrites it at a 6th-grade reading level in the patient's language, and builds a day-by-day plan with reminders. Every generated sentence is linked to the exact source line in the original document, so nothing is invented, and a safety layer checks the result before the patient sees it.

Unique features

- **Source-grounded output:** tap any instruction to highlight the original sentence it came from. Anything the model cannot ground is dropped and flagged for a nurse.
- **Medication safety check:** reconciles the new medicine list against the home list, and flags duplicates, drug-drug interactions and allergy conflicts using RxNorm and openFDA data.
- **Teach-back mode:** short voice or tap questions ("What will you do if your weight goes up 3 lbs?") confirm the patient truly understood. Gaps are flagged for follow-up.
- **Traffic-light symptom guide:** green (normal), amber (call the clinic), red (call 911), tied to the patient's own diagnosis.
- **Caregiver view and multilingual voice:** a family member gets the same plan and reminders; text-to-speech in English, Spanish and Hindi at launch.
- **Clinician review queue:** a nurse sees only the flagged items, so review takes seconds instead of a full re-explanation.

3. Technology Stack and How It Works

Layer	Technology and role
Frontend	React (mobile-first PWA), Tailwind CSS; Web Speech API for voice input and read-aloud
Document intake	PDF text extraction (pdfplumber) with OCR fallback (Tesseract) for photos of paper summaries
Extraction and rewriting	LLM (Claude API) with a strict JSON schema; each field must carry a *source_span* pointing to the original text
Verification layer	Rule-based grounding check (reject any claim without a matching span); drug checks via RxNav/RxNorm and openFDA APIs
Backend	FastAPI (Python), PostgreSQL; encrypted storage, no data retained beyond the session unless the user opts in
Reminders	Web push notifications / SMS (Twilio); calendar (.ics) export for appointments
Data for demo	Synthetic discharge summaries (generated and hand-checked); no real patient data is used

Pipeline: Upload → OCR/text extraction → LLM structured extraction with source spans → grounding and drug-safety verification → plain-language plan in the chosen language → patient/caregiver app with reminders and teach-back → flagged items to the nurse queue.

4. Real-World Impact and Feasibility

- **Impact:** fewer avoidable readmissions and medication errors, better health equity for low-literacy and non-English-speaking patients, and time saved for nurses. Even a small drop in readmissions saves hospitals significant cost.
- **Responsible AI:** it supports clinicians rather than replacing them. It never gives new medical advice beyond the discharge document, shows its sources, uses clear disclaimers and escalates uncertainty to humans.
- **Feasibility:** all components use mature, available APIs and open data. The Round 2 MVP focuses on one condition (heart failure) with 3 synthetic patients, enough for a complete, convincing demo.
- **Scalability and path to adoption:** condition templates can be added incrementally; the structured output can map to FHIR (MedicationStatement, CarePlan) for later EHR integration; privacy design is aligned with HIPAA principles.

5. Round 2 Plan (MVP) and Judging Fit

Criterion	How Discharge Compass addresses it
Innovation and Creativity	Verification-first design with source-linked output and teach-back, instead of another summarizer
Impact and Relevance	Targets a costly, well-documented problem: post-discharge confusion and readmissions
Technical Implementation	Structured extraction, grounding checks, drug-safety APIs, multilingual voice, reminders
UX/UI Design	Large text, traffic-light guide, one-tap source highlight, designed for older adults and caregivers
Presentation	Live demo: upload a discharge summary, get the plan, tap a source, hear Spanish audio, catch a drug conflict

Build plan: Day 1 – extraction and grounding pipeline, drug-check service, core UI. Day 2 – teach-back, voice, reminders, nurse queue, demo polish. Open-source tools and APIs will be attributed as per the rules.